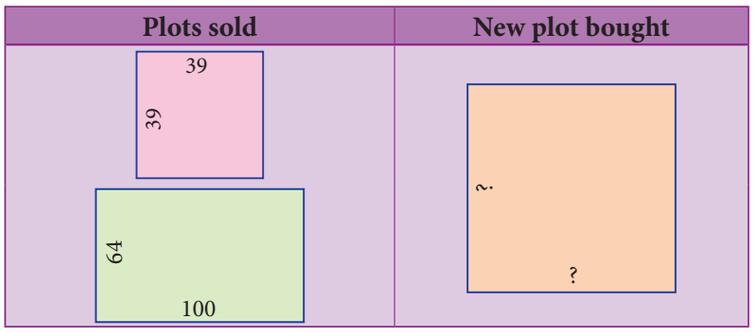
Example 1.27

A real estate owner had two plots, a square plot of side 39 m and a rectangular plot of dimensions 100 m length and 64 m breadth. He sells both of these plots and acquires a new square plot of the same area. What is the length of side of his new plot?

Solution:

The transactions can be visualised as follows:



$$\begin{aligned}
\text{Area of the square plot bought} &= \text{Area of the square plot sold} + \text{Area of the rectangular plot sold} \\
&= 39 \times 39 + 100 \times 64 \\
&= 1521 + 6400 \\
&= 7921 \text{ m}^2
\end{aligned}$$

$$\begin{array}{r|l} & 8\ \ 9 \\ \hline 8 & \overline{79}\ \ \overline{21} \\ & 64 \\ \hline 169 & 15\ \ 21 \\ & 15\ \ 21 \\ \hline & 0 \end{array}$$

Length of a side of the new square plot = $\sqrt{7921}$ = 89 m

Try these

Find the square root by long division method: 1. 400 2. 1764 3. 9801

1.7.3 Number of digits in the square root of a perfect square number

We made use of bars to find the square root in the division method. This marking bars help us to find the number of digits in the square root of a perfect square number. Observe the following examples (with bars shown as if we compute square root by division procedure).

Square root of numbers	No. of bars	No. of digits in the square root	Square root of numbers	No. of bars	No. of digits in the square root
$\sqrt{\overline{169}} = 13$	2	2	$\sqrt{\overline{4356}} = 66$	2	2
$\sqrt{\overline{441}} = 21$	2	2	$\sqrt{\overline{6084}} = 78$	2	2
$\sqrt{\overline{12544}} = 112$	3	3	$\sqrt{\overline{27225}} = 165$	3	3

Hence, we conclude that the number of bars indicates the number of digits in the square root.

Try these

Without calculating the square root, guess the number of digits in the square root of the following numbers: 1. 14400 2. 390625 3. 100000000

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